Week .2

Course Name : Problem Solving and Testing

Course Code : 10211CS227

Name : M Abhi Charan

Vtu No : VTU28517

1. **Build Array from Permutation**
2. **Shuffle the Array**
3. **Find the Highest Altitude**
4. **Java Dequeue**
5. **Java Hashset**

1. Build Array from Permutation

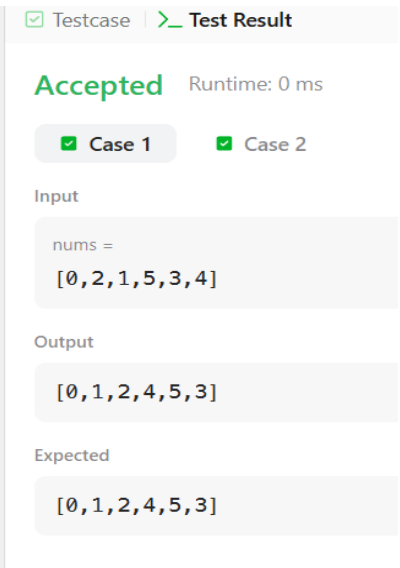
PROGRAM:

```
class Solution {
    public int[] buildArray(int[] nums) {
        int[] ans = new int[nums.length];

        for (int i = 0; i < nums.length; i++) {
            ans[i] = nums[nums[i]];
        }

        return ans;
    }
}
```

OUTPUT:



2. Shuffle the Array

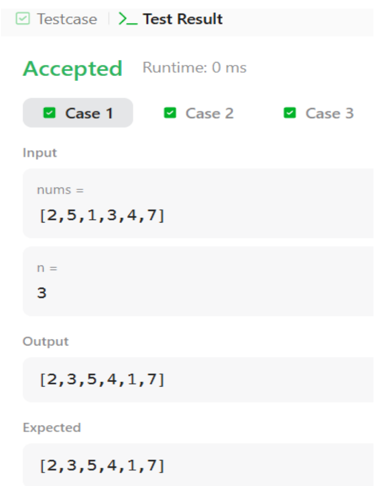
PROGRAM:

```
class Solution {
    public int[] shuffle(int[] nums, int n) {
        int[] ans = new int[2 * n];

        for (int i = 0; i < n; i++) {
            ans[2 * i] = nums[i];
            ans[2 * i + 1] = nums[i + n];
        }

        return ans;
    }
}
```

OUTPUT:



Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | **Case 2** | Case 3

Input

nums =
```
[1,2,3,4,4,3,2,1]
```

n =
```
4
```

Output
```
[1,4,2,3,3,2,4,1]
```

Expected
```
[1,4,2,3,3,2,4,1]
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | **Case 3**

Input

nums =
```
[1,1,2,2]
```

n =
```
2
```

Output
```
[1,2,1,2]
```

Expected
```
[1,2,1,2]
```

3. Find the Highest Altitude

PROGRAM:

```
class Solution {
    public int largestAltitude(int[] gain) {
        int altitude = 0;
        int highest = 0;

        for (int i = 0; i < gain.length; i++) {
            altitude += gain[i];

            if (altitude > highest) {
                highest = altitude;
            }
        }

        return highest;
    }
}
```

OUTPUT:

Accepted Runtime: 0 ms

Case 1 Case 2

Input

gain =

[-5,1,5,0,-7]

Output

1

Expected

1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

gain =

[-4,-3,-2,-1,4,3,2]

Output

0

Expected

0

4. Java Dequeue

PROGRAM:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int m = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        Deque<Integer> deque = new ArrayDeque<>();
        Map<Integer, Integer> freq = new HashMap<>();

        int maxUnique = 0;

        for (int i = 0; i < n; i++) {
            deque.addLast(arr[i]);
```

```
            freq.put(arr[i], freq.getOrDefault(arr[i], 0) + 1);

            // Keep window size equal to m
            if (deque.size() > m) {
                int removed = deque.removeFirst();

                freq.put(removed, freq.get(removed) - 1);

                if (freq.get(removed) == 0) {
                    freq.remove(removed);
                }
            }

            // Number of unique elements in current window
            if (deque.size() == m) {
                maxUnique = Math.max(maxUnique, freq.size());
            }
        }

        System.out.println(maxUnique);
    }
}
```

OUTPUT:

Console I/O AI Agent

```
6 3
5 3 2 6 1 8
3
```

5. Java Hashset

PROGRAM:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        Set<String> set = new HashSet<>();

        for (int i = 0; i < n; i++) {
            String first = sc.next();
            String second = sc.next();

            String pair = first + " " + second;
            set.add(pair);

            System.out.println(set.size());
        }

        sc.close();
    }
}
```

OUTPUT:

Console I/O AI Agent

```
4
SURYA
VAMSHI
1
SIDDHU
SIVA
2
```